TouraLuxe: Software Development Cost & Effort Analysis

1. Executive Summary

This report provides a comprehensive analysis of the TouraLuxe software project, detailing the size of the codebase, estimated development effort, and a market-standard cost analysis. The estimations are based on the industry-standard Constructive Cost Model (COCOMO) and reflect various global market rates for software engineering talent.

2. Codebase Architecture & Size

An automated source code analysis was conducted, excluding third-party dependencies (`node_modules`), build artifacts (`.next`), and version control metadata (`.git`).

Overall Metrics:

- **Total Files:** 109
- **Total Lines of Code (LOC):** 25,073
- **Blank Lines:** 2,361
- **Comment Lines:** 698

Detailed Language Breakdown

Language	Files	Blank	Comment	Code
TypeScript	82	1,800	618	15,876
JSON	4	0	0	7,099
Markdown	4	180	0	622
JavaScript	5	222	28	512
CSS	2	62	14	408
SCSS	1	36	3	196
SQL	2	34	31	152
HTML	2	17	4	148
Text/SVG	7	10	0	60
Total	**109**	**2,361**	**698**	**25,073**

The heavy use of TypeScript (over 63% of the codebase) indicates a modern, type-safe architecture, which generally results in higher maintainability but requires skilled frontend/full-stack engineering talent.

3. Effort & Schedule Estimation

Using the Basic COCOMO model for organic software projects (well-understood requirements, small to medium-sized team), we can mathematically derive the human effort required to build this exact software from scratch.

- **Metric Size (KLOC):** 25.07
- **Estimated Effort:** 70.7 Person-Months
- **Estimated Schedule:** 12.6 Months (Time to develop)
- **Average Team Size:** 5.6 Full-Time Equivalent (FTE) Engineers

Note: A "Person-Month" represents the amount of work performed by one average software engineer in one working month.

4. Market Standard Cost Analysis

Software development costs vary significantly depending on the geographical location and seniority of the engineering team. Below is a breakdown of the estimated project cost (based on 70.7 Person-Months) across three standard global outsourcing tiers.

Tier 1: Onshore (United States / Western Europe)

Typically involves highly integrated, local teams with premium billing rates.

- **Average Monthly Cost per Engineer:** $15,000 - $20,000
- **Estimated Total Project Cost: $1,060,500 - $1,414,000**

Tier 2: Nearshore (Latin America / Eastern Europe)

Offers a balance of time-zone alignment, strong technical skills, and cost efficiency.

- **Average Monthly Cost per Engineer:** $7,000 - $10,000
- **Estimated Total Project Cost: $494,900 - $707,000**

Tier 3: Offshore (India / Southeast Asia)

Provides the highest cost savings, often utilized by large enterprise teams.

- **Average Monthly Cost per Engineer:** $3,500 - $5,000
- **Estimated Total Project Cost: $247,450 - $353,500**

5. Modern Tooling Adjustments

The COCOMO estimations represent traditional development lifecycles. With the advent of modern frameworks (like Next.js/React), vast open-source ecosystems, and AI-assisted development tools (like GitHub Copilot), actual delivery times and costs in today's market can be optimized by **30% to 50%**.

Adjusted Optimized Cost (Global Average Blended Rate at $8,000/month):

- **Traditional Estimate:** $565,600
- **AI-Optimized Estimate (40% efficiency gain): $339,360**

6. Conclusion

TouraLuxe is a substantial mid-sized application featuring over 25,000 lines of code. Replicating this project in the current market would require an estimated **$350,000 to $1,000,000+** depending on team location and the leveraging of AI tooling. The robust use of TypeScript and modern tooling indicates a high-quality codebase that serves as a solid foundation for enterprise scalability.